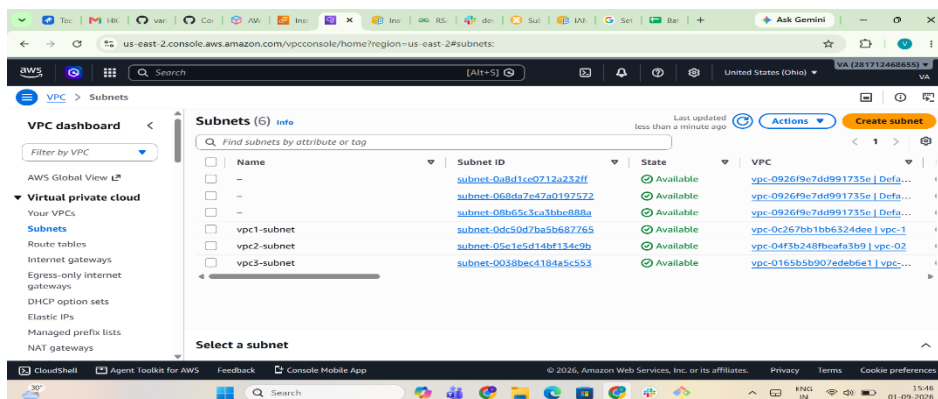
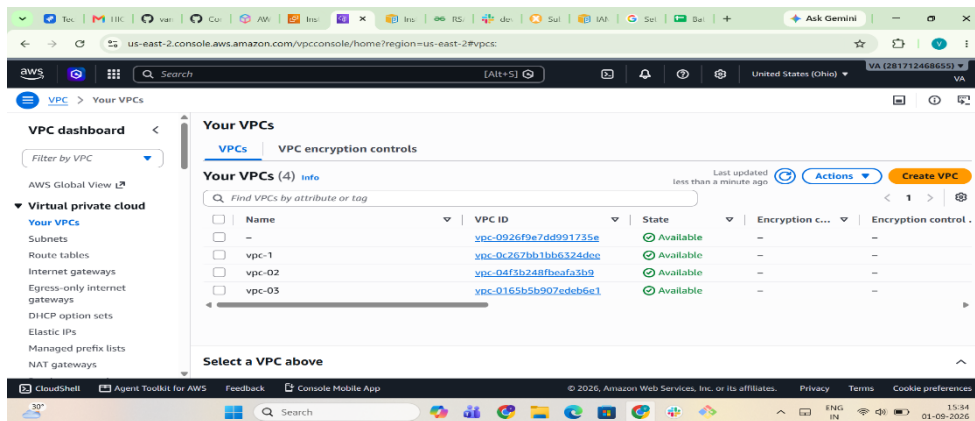


VPC Transit Gateway & VPC Endpoint challenges

1. Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs.

Implementation:

1. Created four VPCs with separate CIDR blocks to avoid IP address conflicts.
2. Created public and private subnets as required in each VPC.
3. Created an AWS Transit Gateway to provide centralized connectivity between the VPCs.
4. Created Transit Gateway VPC attachments for all four VPCs.
5. Associated the four VPC attachments with the Transit Gateway route table.
6. Configured Transit Gateway routes for communication between the CIDR ranges of all four VPCs.
7. Updated the route tables in each VPC to route traffic destined for other VPCs through the Transit Gateway.
8. Configured security groups to allow the required traffic between the VPCs.
9. Launched EC2 instances in the VPCs for connectivity testing.
10. Tested private communication between EC2 instances across the four VPCs.
11. Verified the Transit Gateway attachments, route tables, and connectivity status.



us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#Instances?v=3&instanceState=running

EC2 > Instances

Instances (1/4) Info

Instance state = running

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
vpc02-ec2	i-0e2125771071ecb7f	Running	t3.micro	3/3 checks passed	us-east-2c
vpc03-ec2	i-0593ea340881aef3e	Running	t3.micro	3/3 checks passed	us-east-2c
vpc1-ec2	i-0cf32f104ad660c77	Running	t3.micro	3/3 checks passed	us-east-2a
default-vpc-ec2	i-0961b52fc076165b6	Running	t3.micro	3/3 checks passed	us-east-2b

i-0961b52fc076165b6 (default-vpc-ec2)

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#TransitGatewayAttachments?v=3&case=false%5Cclient:false%5Cregex=tag...

VPC > Transit gateway attachments

Transit gateway attachments (4) Info

Name	Transit gateway attachment ID	Transit gateway ID	State	Resource
default-tga	tgw-attach-056d908543affc2f4	tgw-04ccce342b59c8131	Available	VPC
vpc1-tga	tgw-attach-070a1a090fc051d43	tgw-04ccce342b59c8131	Available	VPC
vpc02-tga	tgw-attach-077ef950baa3befb5	tgw-04ccce342b59c8131	Available	VPC
vpc03-tga	tgw-attach-0d1bbc646c835ee0d	tgw-04ccce342b59c8131	Available	VPC

Select a transit gateway attachment

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#TransitGatewayRouteTableDetails:transitGatewayRouteTableId=tgw-rtb-0d24...

VPC > Transit gateway route tables > tgw-rtb-0d2445e5c1217bb52

Transit gateway ID: tgw-04ccce342b59c8131

Associations (4) Info

Attachment ID	Attachment name	Resource type	Resource ID	Status
tgw-attach-0d1bbc646c835ee0d	vpc03-tga	VPC	vpc-0165b5b907edeb6e1	Available
tgw-attach-056d908543affc2f4	default-tga	VPC	vpc-0926f9e7dd991735e	Available
tgw-attach-077ef950baa3befb5	vpc02-tga	VPC	vpc-04f3b248fbaafa3b9	Available
tgw-attach-070a1a090fc051d43	vpc1-tga	VPC	vpc-0c267bb1bb6324dee	Available

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#EditRoutes:RouteTableId=rtb-0e0b13837a06464cc

Search [Alt+S] Ask Amazon Q United States (Ohio) VA (281712468655)

VPC > Route tables > rtb-0e0b13837a06464cc > Edit routes

Edit routes: rtb-0e0b13837a06464cc / vpc03-RT

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/28	local	Active	No	CreateRouteTable
172.31.0.0/16	Transit Gateway	-	No	CreateRoute
10.0.0.0/28	Transit Gateway	-	No	CreateRoute
10.0.1.0/28	Transit Gateway	-	No	CreateRoute

Add route

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us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#RouteTableDetails:RouteTableId=rtb-0e0b13837a06464cc

Search [Alt+S] Ask Amazon Q United States (Ohio) VA (281712468655)

VPC > Route tables > rtb-0e0b13837a06464cc

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

rtb-0e0b13837a06464cc

Yes

VPC vpc-0165b5b907ede6e1 | vpc-03

Owner ID 281712468655

Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (4)

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/28	tgw-04ccee342b59c8...	Active	No	Create Route
10.0.1.0/28	tgw-04ccee342b59c8...	Active	No	Create Route
10.0.2.0/28	local	Active	No	Create Route Table
172.31.0.0/16	tgw-04ccee342b59c8...	Active	No	Create Route

Both Edit routes

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us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#EditRoutes:RouteTableId=rtb-0712ab238225e58c2

Search [Alt+S] Ask Amazon Q United States (Ohio) VA (281712468655)

VPC > Route tables > rtb-0712ab238225e58c2 > Edit routes

Edit routes: rtb-0712ab238225e58c2 / vpc02-RT

Destination	Target	Status	Propagated	Route Origin
10.0.1.0/28	local	Active	No	CreateRouteTable
172.31.0.0/16	Transit Gateway	-	No	CreateRoute
10.0.0.0/28	Transit Gateway	-	No	CreateRoute
10.0.2.0/28	Transit Gateway	-	No	CreateRoute

Add route

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us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#RouteTableDetails:RouteTableId=rtb-0712ab238225e58c2

VPC dashboard

Route table ID: rtb-0712ab238225e58c2

VPC: vpc-04f3b248fba3b9 | vpc-02

Main: Yes

Owner ID: 281712468655

Explicit subnet associations: -

Edge associations: -

Routes (4)

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/28	tgw-04ccee342b59c8...	Active	No	Create Route
10.0.1.0/28	local	Active	No	Create Route Table
10.0.2.0/28	tgw-04ccee342b59c8...	Active	No	Create Route
172.31.0.0/16	tgw-04ccee342b59c8...	Active	No	Create Route

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#EditRoutes:RouteTableId=rtb-08c5864fd53cb3e9a

Edit routes: rtb-08c5864fd53cb3e9a / vpc1-RT

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/28	local	Active	No	CreateRouteTable
172.31.0.0/16	Transit Gateway	-	No	CreateRoute
10.0.1.0/28	tgw-04ccee342b59c8131	-	No	CreateRoute
10.0.2.0/28	tgw-04ccee342b59c8131	-	No	CreateRoute

Add route

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#RouteTableDetails:RouteTableId=rtb-08c5864fd53cb3e9a

VPC dashboard

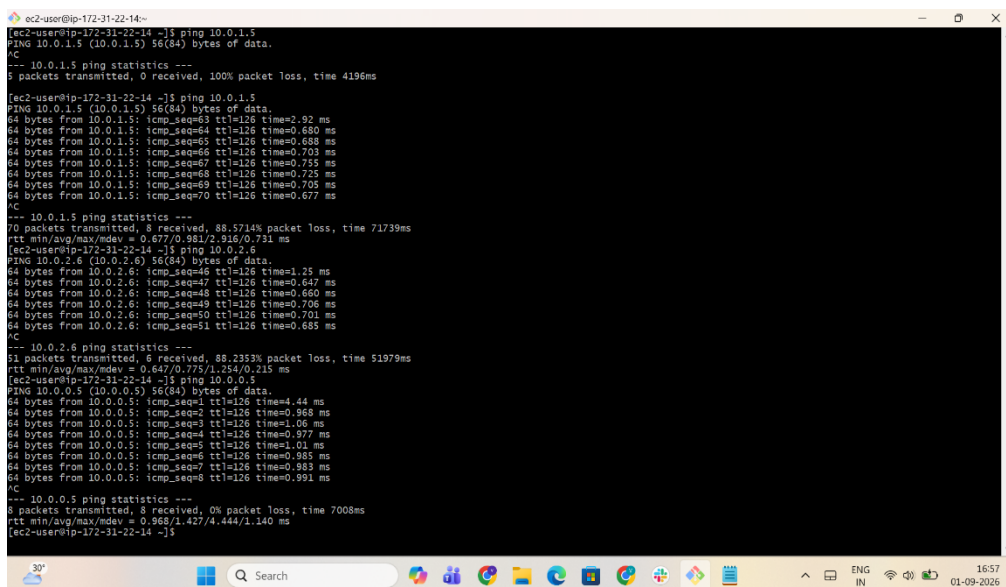
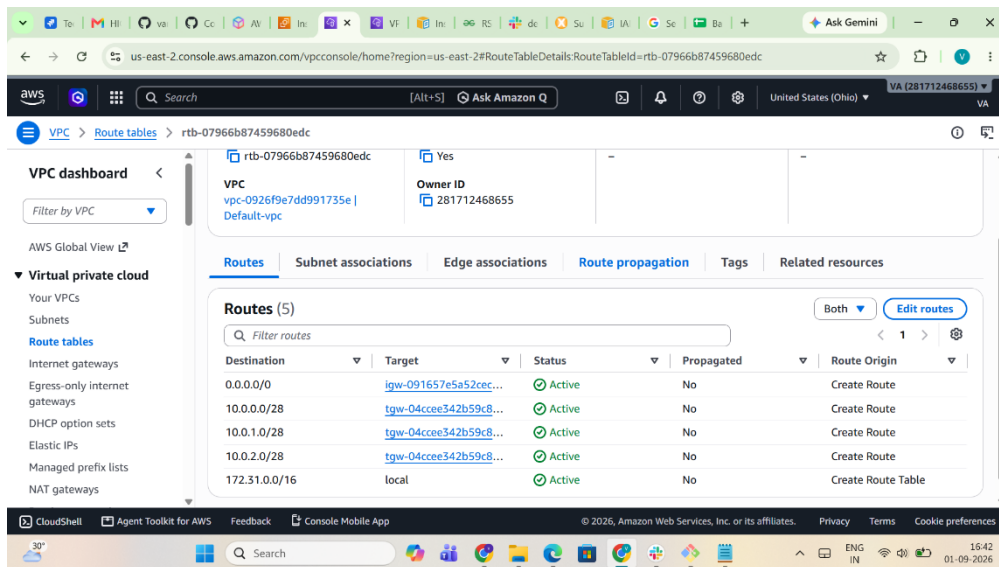
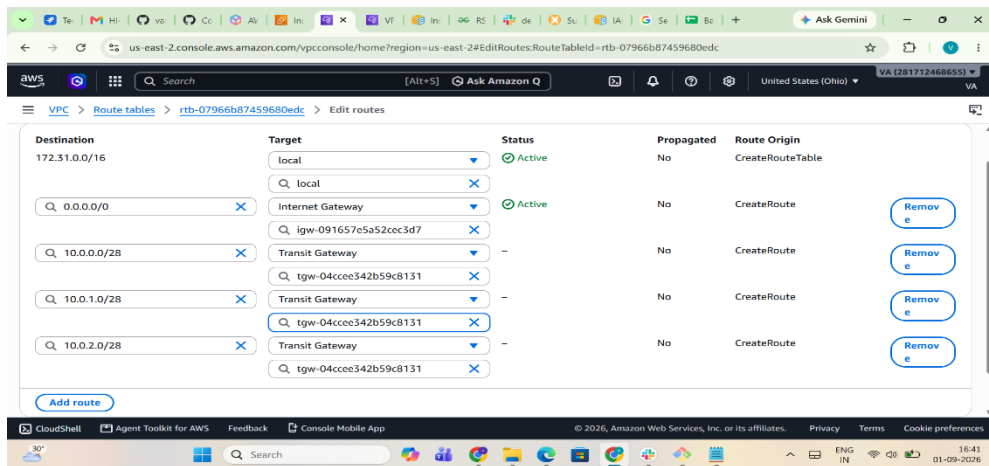
Updated routes for rtb-08c5864fd53cb3e9a / vpc1-RT successfully

VPC: vpc-0c267bb1bb6324dee | vpc-1

Owner ID: 281712468655

Routes (4)

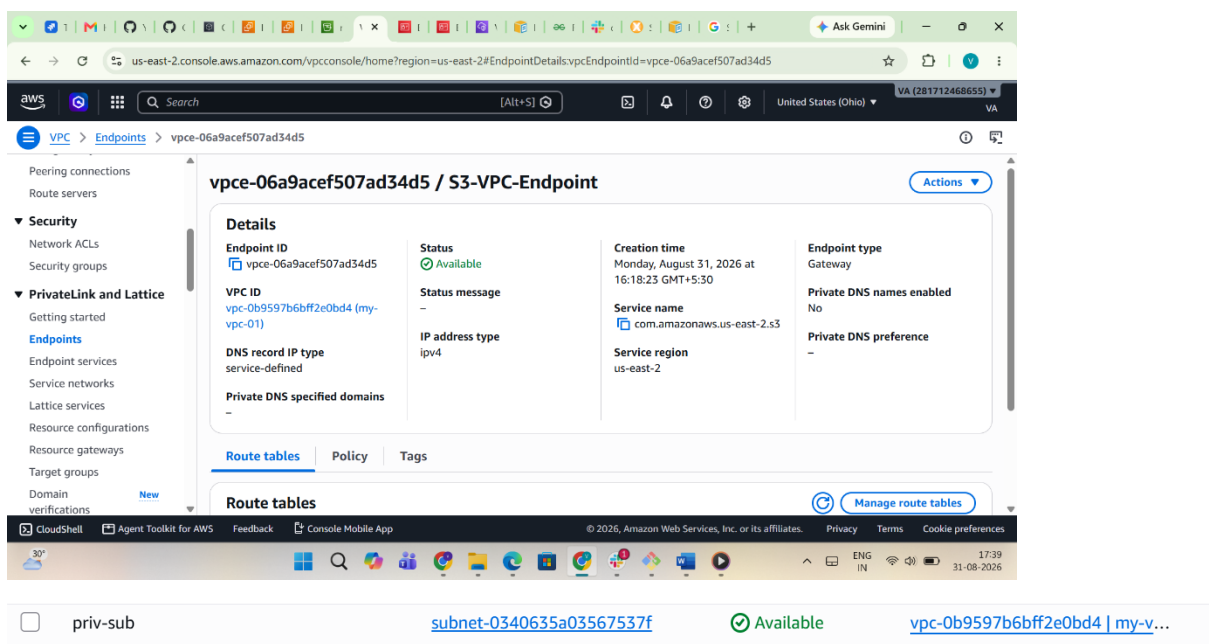
Destination	Target	Status	Propagated	Route Origin
10.0.0.0/28	local	Active	No	Create Route Table
10.0.1.0/28	tgw-04ccee342b59c8...	Active	No	Create Route
10.0.2.0/28	tgw-04ccee342b59c8...	Active	No	Create Route
172.31.0.0/16	tgw-04ccee342b59c8...	Active	No	Create Route



2. Configure VPC endpoints to securely access AWS services without Internet Gateways or NAT Gateways.

Implementation:

1. Used one of the VPCs to configure an S3 Gateway VPC Endpoint.
2. Created an S3 bucket in the same AWS Region for endpoint connectivity testing.
3. Created an S3 Gateway VPC Endpoint for the selected VPC.
4. Associated the S3 VPC Endpoint with the private route table.
5. Verified that the S3 prefix-list route was added to the private route table with the VPC Endpoint as the target.
6. Attached an IAM role to the private EC2 instance with the required S3 permissions.
7. Connected to the private EC2 instance through the available private network path.
8. Created a test file on the private EC2 instance.
9. Used AWS CLI to upload the test file to the S3 bucket through the VPC Endpoint.
10. Verified the uploaded file in the S3 bucket.
11. Confirmed that the private EC2 could access S3 without requiring an Internet Gateway or NAT Gateway.
12. Verified the VPC Endpoint status and route-table association in the AWS Console



us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#RouteTableDetails:RouteTableId=rtb-0a772fd90dad4114

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Details

Route table ID: rtb-0a772fd90dad4114

VPC: vpc-0b9597b6bf2e0bd4 | my-vpc-01

Main: No

Owner ID: 281712468655

Explicit subnet associations: -

Edge associations: -

Routes (2)

Destination	Target	Status	Propagated	Route Origin
pl-7ba54012	vpc-06a9acef507ad3...	Active	No	Create Route
172.168.0.0/24	local	Active	No	Create Route Table

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#InstanceDetails:instanceId=i-037ff4830ceb28bce

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Instance summary for i-037ff4830ceb28bce (ec2-server2)

Updated less than a minute ago

Instance ID: i-037ff4830ceb28bce

Public IPv4 address: 18.226.87.152 | open address

Private IPv4 addresses: 172.168.0.14

Instance state: Running

Private IP DNS name (IPv4 only): ip-172-168-0-14.us-east-2.compute.internal

Public DNS: -

IPV6 address: -

Hostname type: IP name: ip-172-168-0-14.us-east-2.compute.internal

Answer private resource DNS name: -

Auto-assigned IP address: 18.226.87.152 [Public IP]

Instance type: t3.micro

VPC ID: vpc-0b9597b6bf2e0bd4 (my-vpc-01)

Elastic IP addresses: -

AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendations.

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#InstanceDetails:instanceId=i-09a5b2cc4d3ef8296

EC2

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AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Instance summary for i-09a5b2cc4d3ef8296 (my-priv-ec2)

Updated less than a minute ago

Instance ID: i-09a5b2cc4d3ef8296

Public IPv4 address: 16.59.54.82 | open address

Private IPv4 addresses: 172.168.0.24

Instance state: Running

Private IP DNS name (IPv4 only): ip-172-168-0-24.us-east-2.compute.internal

Public DNS: -

IPV6 address: -

Hostname type: IP name: ip-172-168-0-24.us-east-2.compute.internal

Answer private resource DNS name: -

Auto-assigned IP address: 16.59.54.82 [Public IP]

Instance type: t3.micro

VPC ID: vpc-0b9597b6bf2e0bd4 (my-vpc-01)

Elastic IP addresses: -

AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendations.

us-east-1.console.aws.amazon.com/iam/home?region=us-east-2#/roles/details/EC2-S3-Role

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

Roles

Policies

IAM user groups

Identity providers

Account settings

Root access management

Temporary delegation requests

Access reports

Permissions

Trust relationships

Tags

Last Accessed

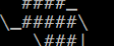
Revoke sessions

Permissions policies (2)

You can attach up to 20 managed policies.

Filter by Type: All types

Policy name	Type	Attached entities
AmazonS3ReadOnlyAccess	AWS managed	1
ec2-s3-policy	Customer inline	0



Amazon Linux 2023

<https://aws.amazon.com/linux/amazon-linux-2023>

```
[ec2-user@ip-172-168-0-24 ~]$ aws sts get-caller-identity
{
  "UserId": "AROAUFD27EKXVBTATC4DN:i-09a5b2cc4d3ef8296",
  "Account": "281712468655",
  "Arn": "arn:aws:sts:281712468655:assumed-role/EC2-S3-Role/i-09a5b2cc4d3ef8296"
}
```

← → ↻ 🏠 us-east-2.console.aws.amazon.com/s3/buckets/my-vpc-endpoint-test-281712468655?region=us-east-2&tab=objects

✔ Successfully created bucket "my-vpc-endpoint-test-281712468655"

my-vpc-endpoint-test-281712468655 [Info](#)

Objects Metadata Properties Permissions Metrics Management File systems - *new* Access Points

Objects (1) Copy S3 URI Copy URL Download Open Delete Actions Create folder Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	test.txt	txt	August 31, 2026, 17:27:45 (UTC+05:30)	24.0 B	Standard